# Project Design Phase-II
# Solution Requirements (Functional & Non-functional)

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| Date | 23 March 2026 |
| Team ID | 69c0f3744241397902eb6826 |
| Project Name | Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking |
| Maximum Marks | 4 Marks |

## ◈ Solution Requirements

## ✅ Functional Requirements:

These describe **what the system should do**:

**1. Data Import Functionality**

- The system should allow users to upload data files (CSV, Excel, APIs).
- It should create **Import Sets** automatically.

**2. Data Transformation**

- The system should transform imported data using **Transform Maps**.
- Field mapping should be configurable.

**3. Relationship Mapping**

- The system should support **Dot Walking** to link related tables.
- It should map relationships (e.g., User → Company → Asset).

**4. Data Validation**

- The system should validate incoming data before processing.
- It should identify missing or incorrect values.

**5. Error Handling**

- The system should log errors during import and transformation.
- Users should be able to view error reports.

**6. Data Storage**

- The system should store processed data in target tables.
- Data integrity must be maintained.

**7. Reporting**

- The system should generate reports after successful import.
- Users should be able to view structured data outputs.

## ⚙ Non-Functional Requirements:

These describe **how the system should perform**:

**1. Performance**

- Data import should be fast and efficient.
- System should handle large datasets without delay.

**2. Scalability**

- The system should support increasing data volume.
- It should scale with additional users and records.

**3. Reliability**

- The system should ensure accurate data processing.
- Failures should not result in data loss.

**4. Security**

- Only authorized users should upload/import data.
- Data should be protected from unauthorized access.

**5. Usability**

- The interface should be user-friendly.
- Minimal technical knowledge should be required.

**6. Maintainability**

- The system should be easy to update and modify.
- Transform maps and mappings should be editable.

**7. Availability**

- The system should be available with minimal downtime.
- Backup and recovery mechanisms should be in place.

## 📚 Real-Time Examples (ServiceNow)

### 🔷 1. Employee Data Import (HR Module)

#### 📌 Scenario

A company uploads a **CSV file of new employees** into ServiceNow.

#### ⚙️ How it works

- File uploaded → **Import Set Table (u_employee_import)**
- Transform Map maps fields → Name, Email, Department
- **Dot Walking** used:
  - department.name → Links employee to Department table

#### ✅ Outcome

- Employees automatically created
- Correct department assigned

### 🔷 2. Incident Assignment Automation (ITSM)

#### 📌 Scenario

Incidents are imported from an external monitoring tool.

#### ⚙️ Process

- Data comes via API → Import Set
- Transform Map maps → Short description, Priority
- **Dot Walking**:
  - assignment_group.name → Maps to correct support team

### ✅ Outcome

- Incidents auto-assigned to right team
- No manual intervention needed

## 🔷 3. Asset Management Integration

### 📌 Scenario

Hardware assets (laptops, servers) imported from Excel.

### ⚙️ Process

- Excel upload → Import Set
- Transform Map → Serial number, Model, Owner
- **Dot Walking**:
  - owner.email → Links asset to user record
  - model.name → Links to model table

### ✅ Outcome

- Assets correctly linked to users
- Accurate asset tracking